DDA3020 Machine Learning Tutorial: Principal Component Analysis

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Instructions

This exercise sheet covers the main ideas of Principal Component Analysis (PCA), including: the projection of data points onto a low-dimensional subspace and the orthogonal theorem; the two equivalent optimization objectives of PCA (maximizing the projected variance vs. minimizing the reconstruction error); the necessity of data centering; the formulation of the empirical covariance matrix; the derivation of the optimal solution via Lagrange multipliers; the role of eigen-decomposition/SVD in solving the PCA problem; the interpretation of eigenvalues as the variance of each principal component; the variance explained ratio and the selection of the number of principal components; the reconstruction error and its relation to the discarded eigenvalues; the decorrelation property of the PCA output; and the standard PCA algorithm steps. You should be able to derive the optimal solution of PCA from scratch, to compute the principal components for small datasets, and to explain the key properties of PCA.

Useful Formulas

1. Projection and reconstruction. Given an orthonormal basis matrix $U \in \mathbb{R}^{D\times K}$ ($U^\top U = I$) and data mean μ, the low-dimensional representation of $\mathbf{x}$ is:

   $$\mathbf{z} = U^\top(\mathbf{x} - \mu)$$

   The reconstruction of $\mathbf{x}$ from $\mathbf{z}$ is:

   $$\tilde{\mathbf{x}} = \mu + U\mathbf{z}$$

2. Orthogonal theorem. The reconstruction residual is orthogonal to the subspace:

   $$U^\top(\mathbf{x} - \tilde{\mathbf{x}}) = 0$$

3. Empirical covariance matrix. For a centered dataset (i.e., $\mu = 0$ after centering), the empirical covariance matrix is:

   $$\Sigma = \frac{1}{N}\sum_{n=1}^{N} \mathbf{x}^{(n)}(\mathbf{x}^{(n)})^\top$$

4. Eigen-decomposition of covariance matrix. The covariance matrix can be decomposed as:

   $$\Sigma = Q\Lambda Q^\top, \quad \Lambda = \mathrm{diag}(\lambda_1, \lambda_2, ..., \lambda_D), \quad \lambda_1 \geq \lambda_2 \geq ... \geq \lambda_D$$

   where $\mathbf{q}_i$ (the i-th column of Q) is the eigenvector corresponding to eigenvalue λ_i.

5. Variance explained ratio. The ratio of variance preserved by the top-K principal components is:

$$r_K = \frac{\sum_{i=1}^{K} \lambda_i}{\sum_{i=1}^{D} \lambda_i}$$

6. Reconstruction error. The average reconstruction error when using top-K principal components is:

$$\frac{1}{N} \sum_{n=1}^{N} \|\mathbf{x}^{(n)} - \tilde{\mathbf{x}}^{(n)}\|^2 = \sum_{i=K+1}^{D} \lambda_i$$

7. Equivalence of objectives. The two PCA objectives are equivalent due to the Pythagorean theorem:

$$\frac{1}{N} \sum_{n} \|\tilde{\mathbf{x}}^{(n)} - \mu\|^2 + \frac{1}{N} \sum_{n} \|\mathbf{x}^{(n)} - \tilde{\mathbf{x}}^{(n)}\|^2 = \frac{1}{N} \sum_{n} \|\mathbf{x}^{(n)} - \mu\|^2 = \text{constant}$$

Additional Exercises

1. (Orthogonality of reconstruction residual). Consider the projection of a data point $\mathbf{x}$ onto a K-dimensional subspace spanned by orthonormal basis $U = [\mathbf{u}_1, ..., \mathbf{u}_K]$. Let $\tilde{\mathbf{x}}$ be the reconstruction of $\mathbf{x}$ after projection. Prove that the residual $\mathbf{x} - \tilde{\mathbf{x}}$ is orthogonal to the subspace, i.e., $U^\top(\mathbf{x} - \tilde{\mathbf{x}}) = 0$.

2. (Equivalence of PCA objectives). PCA can be derived from two different perspectives: maximizing the variance of the projected data, or minimizing the reconstruction error. Prove that these two objectives are equivalent. Specifically, show that maximizing $\frac{1}{N}\sum_n \|\tilde{\mathbf{x}}^{(n)} - \mu\|^2$ is equivalent to minimizing $\frac{1}{N}\sum_n \|\mathbf{x}^{(n)} - \tilde{\mathbf{x}}^{(n)}\|^2$, given that the total variance $\frac{1}{N}\sum_n \|\mathbf{x}^{(n)} - \mu\|^2$ is a constant independent of U.

3. (Necessity of data centering). A common preprocessing step for PCA is to center the data by subtracting the mean μ from each data point. Explain why this centering step is necessary for PCA.

4. (PCA on a small dataset). Consider the 2-dimensional dataset with 5 points:

$$X = \begin{pmatrix} -1 & -1 & 0 & 2 & 0 \\ -2 & 0 & 0 & 1 & 1 \end{pmatrix}$$

You may assume the data is already centered (i.e., $\mu = 0$).

   (a) Compute the empirical covariance matrix Σ of this dataset.
   (b) Compute the eigenvalues and eigenvectors of Σ.
   (c) If we reduce the dimensionality to $K = 1$, what is the projection matrix U?
   (d) Compute the low-dimensional representation $\mathbf{z}^{(n)}$ for each data point.

5. (Decorrelation of PCA features). Prove that after applying PCA, the dimensions of the low-dimensional representation $\mathbf{z}$ are uncorrelated. Specifically, show that the covariance matrix of $\mathbf{z}$ is a diagonal matrix.

6. (Reconstruction error from eigenvalues). Suppose we have a dataset whose covariance matrix has eigenvalues $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_D$. Prove that when we use the top-K principal

components for reconstruction, the average reconstruction error equals the sum of the discarded eigenvalues:

$$\frac{1}{N}\sum_{n=1}^{N}\|\mathbf{x}^{(n)}-\tilde{\mathbf{x}}^{(n)}\|^2=\sum_{i=K+1}^{D}\lambda_i$$

7. (Selecting the number of principal components). Suppose we have a 7-dimensional dataset, and the eigenvalues of its covariance matrix are:

$$\lambda=[5.0,3.0,2.0,1.0,0.5,0.3,0.2]$$

We want to retain at least 90% of the variance of the original data. What is the minimum number of principal components we need to choose? Show your calculation.

8. (Deriving PCA via Lagrange multipliers). Consider the case of finding the first principal component ($K=1$). The optimization problem is:

$$\max_{\mathbf{u}}\mathbf{u}^\top\Sigma\mathbf{u}\quad\text{s.t.}\quad\mathbf{u}^\top\mathbf{u}=1$$

where Σ is the empirical covariance matrix. Using the method of Lagrange multipliers, derive that the optimal solution $\mathbf{u}^*$ must be an eigenvector of Σ, and the corresponding eigenvalue is the maximum value of the objective function.

9. (PCA vs. Linear Regression). In 2-dimensional data, we often fit a line to the data points.

   - Linear regression fits the line $y=ax+b$ by minimizing the vertical residual sum of squares: $\sum_n(y^{(n)}-(ax^{(n)}+b))^2$.
   - PCA fits the line by minimizing the orthogonal residual sum of squares: $\sum_n\|\mathbf{x}^{(n)}-\tilde{\mathbf{x}}^{(n)}\|^2$, which is exactly the reconstruction error.

   Explain the key difference between these two methods, and why they can produce very different lines even for the same dataset.

10. (Variance of projected data). Prove that the variance of the data projected onto the eigenvector $\mathbf{q}_i$ of the covariance matrix is exactly the corresponding eigenvalue λ_i.

11. (Properties of reconstructed data). Suppose we run PCA on the original dataset X to get the reconstructed dataset $\tilde{X}$. Now, if we run PCA again on $\tilde{X}$ to reduce it to K dimensions, will we get the same projection matrix as before? Explain your answer, and describe what the principal components of $\tilde{X}$ would be.

Solutions

1. **Solution to Q1 (Orthogonality of reconstruction residual).** By the definition of reconstruction, we have:
$$\mathbf{x} - \tilde{\mathbf{x}} = \mathbf{x} - \mu - U\mathbf{z}$$
And by the definition of $\mathbf{z}$, we have $\mathbf{z} = U^\top(\mathbf{x} - \mu)$. Substituting this into the above equation:
$$U^\top(\mathbf{x} - \tilde{\mathbf{x}}) = U^\top(\mathbf{x} - \mu) - U^\top U\mathbf{z}$$
Since U is orthonormal, $U^\top U = I$. Therefore:
$$U^\top(\mathbf{x} - \tilde{\mathbf{x}}) = \mathbf{z} - I\mathbf{z} = 0$$
This completes the proof.

2. **Solution to Q2 (Equivalence of PCA objectives).** From the Pythagorean theorem, for each data point, we have:
$$\|\mathbf{x}^{(n)} - \mu\|^2 = \|\tilde{\mathbf{x}}^{(n)} - \mu\|^2 + \|\mathbf{x}^{(n)} - \tilde{\mathbf{x}}^{(n)}\|^2$$
This holds because the residual $\mathbf{x}^{(n)} - \tilde{\mathbf{x}}^{(n)}$ is orthogonal to the subspace, so the three points $\mu, \tilde{\mathbf{x}}^{(n)}, \mathbf{x}^{(n)}$ form a right triangle. Averaging over all data points, we get:
$$\frac{1}{N}\sum_n \|\mathbf{x}^{(n)} - \mu\|^2 = \frac{1}{N}\sum_n \|\tilde{\mathbf{x}}^{(n)} - \mu\|^2 + \frac{1}{N}\sum_n \|\mathbf{x}^{(n)} - \tilde{\mathbf{x}}^{(n)}\|^2$$
The left-hand side is the total variance of the original data, which is a constant independent of the choice of subspace U. Therefore, maximizing the first term on the right-hand side (the projected variance) is exactly equivalent to minimizing the second term on the right-hand side (the reconstruction error).

3. **Solution to Q3 (Necessity of data centering).** The centering step is necessary because PCA aims to capture the variance of the data around the mean. The covariance matrix, which is the core of PCA, is defined based on the deviations from the mean. Without centering, the objective function of PCA would be affected by the mean of the data, leading to incorrect principal components that are biased towards the mean position rather than the direction of maximum variance.

4. **Solution to Q4 (PCA on a small dataset).**

   (a) The empirical covariance matrix is:
$$\Sigma = \frac{1}{5}XX^\top = \frac{1}{5}\begin{pmatrix} -1 & -1 & 0 & 2 & 0 \\ -2 & 0 & 0 & 1 & 1 \end{pmatrix}\begin{pmatrix} -1 & -2 \\ -1 & 0 \\ 0 & 0 \\ 2 & 1 \\ 0 & 1 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 6 & 4 \\ 4 & 6 \end{pmatrix}$$

   (b) To find eigenvalues, we solve $\det(\Sigma - \lambda I) = 0$:
$$\det\left(\begin{pmatrix} 6/5 - \lambda & amp;4/5 \\ 4/5 & amp;6/5 - \lambda \end{pmatrix}\right) = (6/5 - \lambda)^2 - (4/5)^2 = 0$$
This gives $6/5 - \lambda = \pm 4/5$, so $\lambda_1 = 2$, $\lambda_2 = 2/5$. For $\lambda_1 = 2$, the eigenvector satisfies $(\Sigma - 2I)\mathbf{q}_1 = 0$, which gives $\mathbf{q}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$. For $\lambda_2 = 2/5$, the eigenvector is $\mathbf{q}_2 = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$.

(c) For $K = 1$, we take the top eigenvector as U, so $U = \mathbf{q}_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$.

(d) The low-dimensional representation is $\mathbf{z} = U^\top X$:

$$\mathbf{z} = \begin{pmatrix} 1/\sqrt{2} & amp; 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} -1 & amp; -1 & amp; 0 & amp; 2 & amp; 0 \\ -2 & amp; 0 & amp; 0 & amp; 1 & amp; 1 \end{pmatrix} = \begin{pmatrix} -3/\sqrt{2} & amp; -1/\sqrt{2} & amp$$

5. **Solution to Q5 (Decorrelation of PCA features)**. The covariance matrix of $\mathbf{z}$ is:

$$\text{Cov}(\mathbf{z}) = \text{Cov}(U^\top(\mathbf{x} - \mu)) = U^\top \text{Cov}(\mathbf{x}) U$$

We know that $\text{Cov}(\mathbf{x}) = \Sigma = Q\Lambda Q^\top$, and U is the matrix of the top-K eigenvectors of Σ, so $U = Q_K$ (the first K columns of Q). Substituting:

$$\text{Cov}(\mathbf{z}) = U^\top Q \Lambda Q^\top U$$

Since Q is orthogonal, $Q^\top U = Q^\top Q_K = I_{K\times K}$ (the identity matrix, because the first K columns of Q are orthonormal). Therefore:

$$\text{Cov}(\mathbf{z}) = I_K^\top \Lambda I_K = \Lambda_K$$

where Λ_K is the top-left $K \times K$ block of Λ, which is a diagonal matrix. This means the covariance between different dimensions of $\mathbf{z}$ is zero, so they are uncorrelated.

6. **Solution to Q6 (Reconstruction error from eigenvalues)**. From the equivalence of objectives, we know that:

$$\frac{1}{N}\sum_n \|\mathbf{x}^{(n)} - \tilde{\mathbf{x}}^{(n)}\|^2 = \frac{1}{N}\sum_n \|\mathbf{x}^{(n)} - \mu\|^2 - \frac{1}{N}\sum_n \|\tilde{\mathbf{x}}^{(n)} - \mu\|^2$$

The first term is the total variance of the original data, which equals the sum of all eigenvalues: $\sum_{i=1}^{D} \lambda_i$. The second term is the projected variance, which equals the sum of the top-K eigenvalues: $\sum_{i=1}^{K} \lambda_i$. Therefore, subtracting them gives:

$$\frac{1}{N}\sum_n \|\mathbf{x}^{(n)} - \tilde{\mathbf{x}}^{(n)}\|^2 = \sum_{i=1}^{D} \lambda_i - \sum_{i=1}^{K} \lambda_i = \sum_{i=K+1}^{D} \lambda_i$$

This completes the proof.

7. **Solution to Q7 (Selecting the number of principal components)**. First, compute the total variance:

$$\text{Total} = 5.0 + 3.0 + 2.0 + 1.0 + 0.5 + 0.3 + 0.2 = 12.0$$

Now compute the cumulative variance explained for increasing K:

- $K = 1$: $5.0/12.0 \approx 41.67\%$
- $K = 2$: $(5 + 3)/12 = 8/12 \approx 66.67\%$
- $K = 3$: $(5 + 3 + 2)/12 = 10/12 \approx 83.33\%$
- $K = 4$: $(5 + 3 + 2 + 1)/12 = 11/12 \approx 91.67\%$

We can see that when $K = 4$, the variance explained is 91.67%, which is above 90%. For $K = 3$, it is only 83.33%, which is below 90%. Therefore, the minimum number of principal components we need is 4.

8. **Solution to Q8 (Deriving PCA via Lagrange multipliers)**. We construct the Lagrangian for the optimization problem:

$$L(\mathbf{u}, \lambda) = \mathbf{u}^\top \Sigma \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1)$$

where λ is the Lagrange multiplier. To find the optimum, we take the derivative of L with respect to $\mathbf{u}$ and set it to zero:

$$\frac{\partial L}{\partial \mathbf{u}} = 2\Sigma \mathbf{u} - 2\lambda \mathbf{u} = 0$$

Rearranging, we get:

$$\Sigma \mathbf{u} = \lambda \mathbf{u}$$

This is exactly the definition of an eigenvector: $\mathbf{u}$ must be an eigenvector of Σ, and λ is the corresponding eigenvalue. Now, the objective function is $\mathbf{u}^\top \Sigma \mathbf{u} = \mathbf{u}^\top(\lambda \mathbf{u}) = \lambda \mathbf{u}^\top \mathbf{u} = \lambda$, since $\mathbf{u}^\top \mathbf{u} = 1$. Therefore, the value of the objective function is exactly the eigenvalue. To maximize the objective, we need to choose the largest possible λ, which corresponds to the largest eigenvalue of Σ.

9. **Solution to Q9 (PCA vs. Linear Regression)**. The key difference lies in what error they are minimizing:

   - Linear regression assumes that x is the input variable and y is the output variable. It minimizes the error in the y-direction, i.e., how far the points are from the line vertically. This is asymmetric: it treats x and y differently, with the goal of predicting y from x.
   - PCA treats x and y symmetrically, as two dimensions of the data. It minimizes the orthogonal distance from the points to the line, i.e., the Euclidean distance between the original point and its reconstruction on the line. This is symmetric, with the goal of finding the best lower-dimensional representation of the data, regardless of prediction.

   Because of this difference, they can produce very different lines. For example, if the data has a large variance in the x-direction, linear regression will fit a line that predicts y well, while PCA will fit a line that captures the overall spread of the data, which can be very different.

10. **Solution to Q10 (Variance of projected data)**. The variance of the projected data is:

$$\mathrm{Var}(\mathbf{q}_i^\top \mathbf{x}) = \mathbb{E}[(\mathbf{q}_i^\top \mathbf{x} - \mathbf{q}_i^\top \mu)^2] = \mathbb{E}[\mathbf{q}_i^\top (\mathbf{x} - \mu)(\mathbf{x} - \mu)^\top \mathbf{q}_i]$$

This can be rewritten as:

$$\mathbf{q}_i^\top \mathbb{E}[(\mathbf{x} - \mu)(\mathbf{x} - \mu)^\top] \mathbf{q}_i = \mathbf{q}_i^\top \Sigma \mathbf{q}_i$$

Since $\mathbf{q}_i$ is an eigenvector of Σ, we have $\Sigma \mathbf{q}_i = \lambda_i \mathbf{q}_i$. Substituting:

$$\mathbf{q}_i^\top \Sigma \mathbf{q}_i = \mathbf{q}_i^\top \lambda_i \mathbf{q}_i = \lambda_i \mathbf{q}_i^\top \mathbf{q}_i = \lambda_i$$

because $\mathbf{q}_i$ is a unit vector, so $\mathbf{q}_i^\top \mathbf{q}_i = 1$. Therefore, the variance of the projected data is exactly λ_i.

11. **Solution to Q11 (Properties of reconstructed data)**. Yes, we will get the same projection matrix. The reconstructed data is $\tilde{\mathbf{x}}^{(n)} = \mu + UU^\top(\mathbf{x}^{(n)} - \mu)$, where U is the original projection matrix. The covariance matrix of $\tilde{X}$ is:

$$\tilde{\Sigma} = \frac{1}{N} \sum_n (\tilde{\mathbf{x}}^{(n)} - \mu)(\tilde{\mathbf{x}}^{(n)} - \mu)^\top = UU^\top \Sigma UU^\top$$

We know that $\Sigma U = U\Lambda_K$, where Λ_K is the diagonal matrix of the top-K eigenvalues. Substituting:

$$\tilde{\Sigma} = UU^\top U\Lambda_K U^\top U = U\Lambda_K U^\top$$

because $U^\top U = I$. This means the eigenvectors of $\tilde{\Sigma}$ are exactly the columns of U, with eigenvalues equal to the original top-K eigenvalues. Therefore, when we run PCA on $\tilde{X}$, the top-K eigenvectors are exactly the original U, so the projection matrix is the same as before.